Evidence Report

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Title:	Workstation Image Examination I – Winhex
Case:	26-T005
Date:	04/03/2026

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Document Revision History

Name	Revision Date	Version	Description
Tytus Gonzalez	2026-04-01	0.1	Draft
Tytus Gonzalez	2026-04-03	1.0	Final

Executive Summary

On 4/3/2026, a corporate workstation was reported to have had sensitive documents stolen and posted on a competitor's website. A forensic investigation was conducted using a forensic image of the workstation to determine the partition structure, verify the integrity of the disk, and identify any observable data on the device. I, Tytus Gonzalez, prepared the Windows forensic virtual machine and used WinHex to examine the disk image in a controlled environment.

The investigation began with verifying the integrity of the disk image using the provided hashing tools to ensure no corruption had occurred during acquisition. The image was then opened in WinHex to conduct a low-level analysis of the disk, starting at offset 0x000 to inspect the boot code, disk signature, and partition table. The first 440 bytes were examined to confirm the presence of boot code, and the disk signature at offset 0x01B8 was recorded to ensure the image matched the expected disk configuration. Next, the partition table located at offset 0x01BE was analyzed. Four partition entries were present, including empty entries. Partition 1 was identified as bootable and of type NTFS (hexadecimal value 0x07). The LBA start address of this partition was calculated as 63, corresponding to a byte offset of 32,256. The partition size was determined to be 20,948,697 sectors, equating to 10,725,732,864 bytes, which closely matched the total size of the disk image. The first bytes at the LBA start offset were examined, revealing the NTFS boot sector signature, confirming the file system type and partition integrity. Based on these observations, the disk image uses a Master Boot Record (MBR) partitioning scheme, as evidenced by the standard four-entry partition table, disk signature, and the boot sector signature at the start of Partition 1. No anomalies were observed in the low-level structure of the disk image, indicating the integrity of the partition layout and confirming the NTFS file system for the primary partition.

This investigation demonstrates a thorough forensic analysis process, including verification of disk integrity, examination of boot code and partition tables, calculation of LBA start addresses and partition sizes, and validation of filesystem types. These steps provide a clear understanding of the structure and organization of the workstation's storage, which is essential for further examination.

Synopsis

Review the first 446 bytes

1. Does this image have boot code in the first 440 bytes?

2. Give your observations including any plaintext in the code.

3. What is the "disk signature" starting at offset 0x01B8? Provide the value as a hexadecimal value like 0x01234567; Give the correct value with endianness in consideration

Review the partition table beginning at offset 0x01BE

1. How many partition entries are in the table? Include empty ones.

2. Which partition entry is marked bootable?

3. Partition entry 1 - What is the partition entry type? Give the hexadecimal value and the type (e.g. Linux, FAT32)

4. Partition entry 1 - What is the LBA start address? Give the hexadecimal value with endianness in consideration

5. Calculate the offset of the LBA start address. Sector size is 512 bytes or 0x200 in hexadecimal.

6. Partition entry 1 - What is the partition size? Give the hexadecimal value with endianness in consideration

7. Calculate the size of the partition in bytes. Sector size is 512 bytes or 0x200 in hexadecimal Size must be in decimal

8. What is the difference, in bytes, between the actual size of the evidence file versus the value above?

Navigate to Partition Entry One LBA Start Address Offset

1. What are the first 8 bytes at this offset? DO NOT consider endianness in this value

2. Does this value support your partition type answer above?

3. Give your observations including any plaintext in the code.

4. Based on your observations, does this disk image use an MBR or GPT partitioning scheme? Explain why.

Evidence Collected

The section outlines all physical evidence items collected

Name	Identifier (Tag)	Description
Windows-image-01.dd	Windows-image-01	DD image of Windows machine provided by instructor

This section provides details of the digital evidence collected

Name	Windows-image-01.dd
Type:	DOS/MBR boot sector MS-MBR XP english at offset 0x12c "Invalid partition table" at offset 0x144 "Error loading operating system" at offset 0x163 "Missing operating system", disk signature 0x39bf39be; partition 1 : ID=0x7, active, start-CHS (0x0,1,1), end-CHS (0x3ff,254,63), startsector 63, 20948697 sectors
Size:	10737418240 bytes
MD5:	78a52b5bac78f4e711607707ac0e3f93
SHA1:	ba7dc57e08bb6e3393aee15c713ae04feadcd181
SHA256	4d4fc46f284630a69980340ee36d3ed486de424a9f456eedba0f7194cc7a991b

Tools Used

The following outlines the workstations and all hardware and software used for this investigation.

Workstation

Hostname	Operating System	Build	Physical / Virtual	Built
Windows 11x64_DFIR	Windows 11 Home	2021	Virtual	02/06/2026

Software

Name	Version	Release	Purpose
WinHex/X-ways Forensics	21.6	Oct 19, 2025	A Universal hexadecimal editor, particularly helpful in the realm of computer forensics, data recovery, low-level data processing, and IT security

Case Information & Observables

This case involved collecting, processing, and analyzing a Windows-based corporate workstation provided by the Client. The workstation was reported to have had sensitive corporate documents stolen and posted on a competitor's website. The following are my observations and responses to Client questions regarding the disk image, partition layout, and filesystem structure.

Observables

78A52B5BAC78F4E711607707AC03F93

<u>Review the first 446 bytes</u>

1. Does this image have boot code in the first 440 bytes?

Yes, the first 440 bytes contain boot code. This is evident because the bytes are not all zeros and include executable instructions, as well as readable text strings commonly found in MBR code.

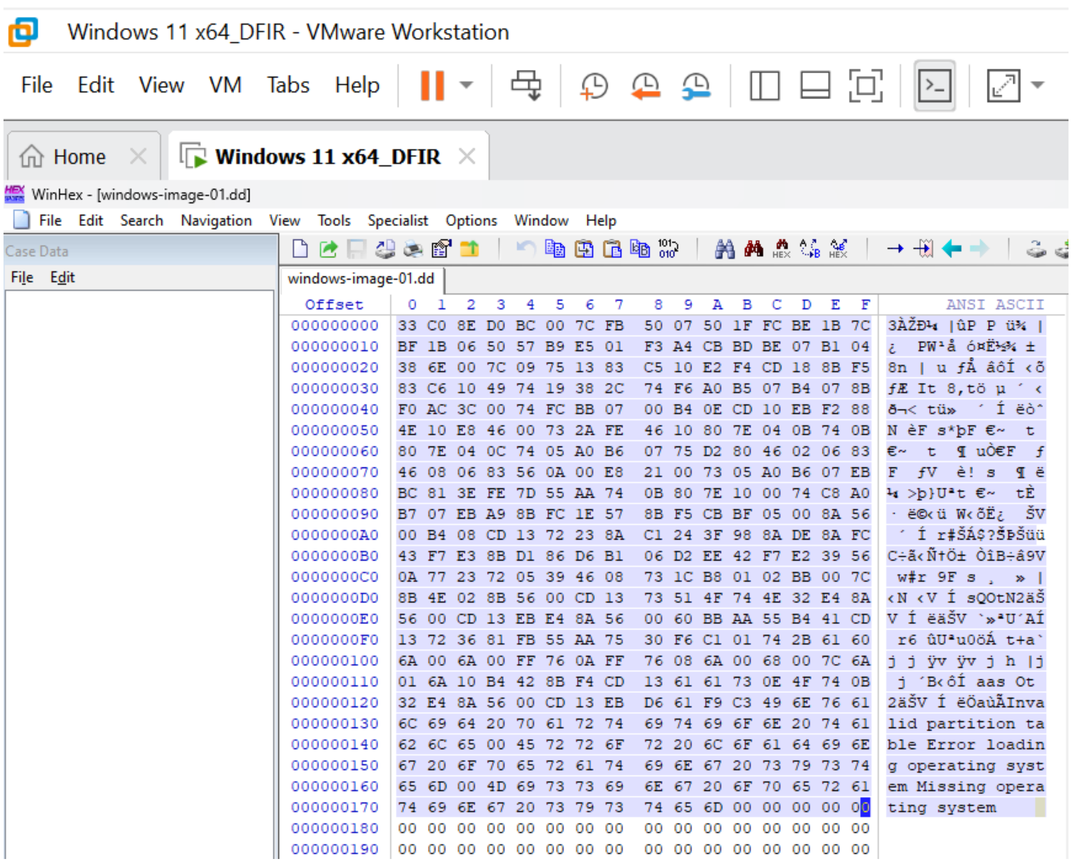

Figure 1: The hexadecimal value of the first 440 bytes of the dd; the plaintext on the side confirms that this is MBR boot code

2. Give your observations including any plaintext in the code.

Within the first 440 bytes, there are two notable plaintext strings: "Invalid partition table", "Missing operating system", and "Error loading operating system". This confirms that the first 440 bytes are of MBR boot code, which is designed to display errors when the system cannot find a valid bootable partition.

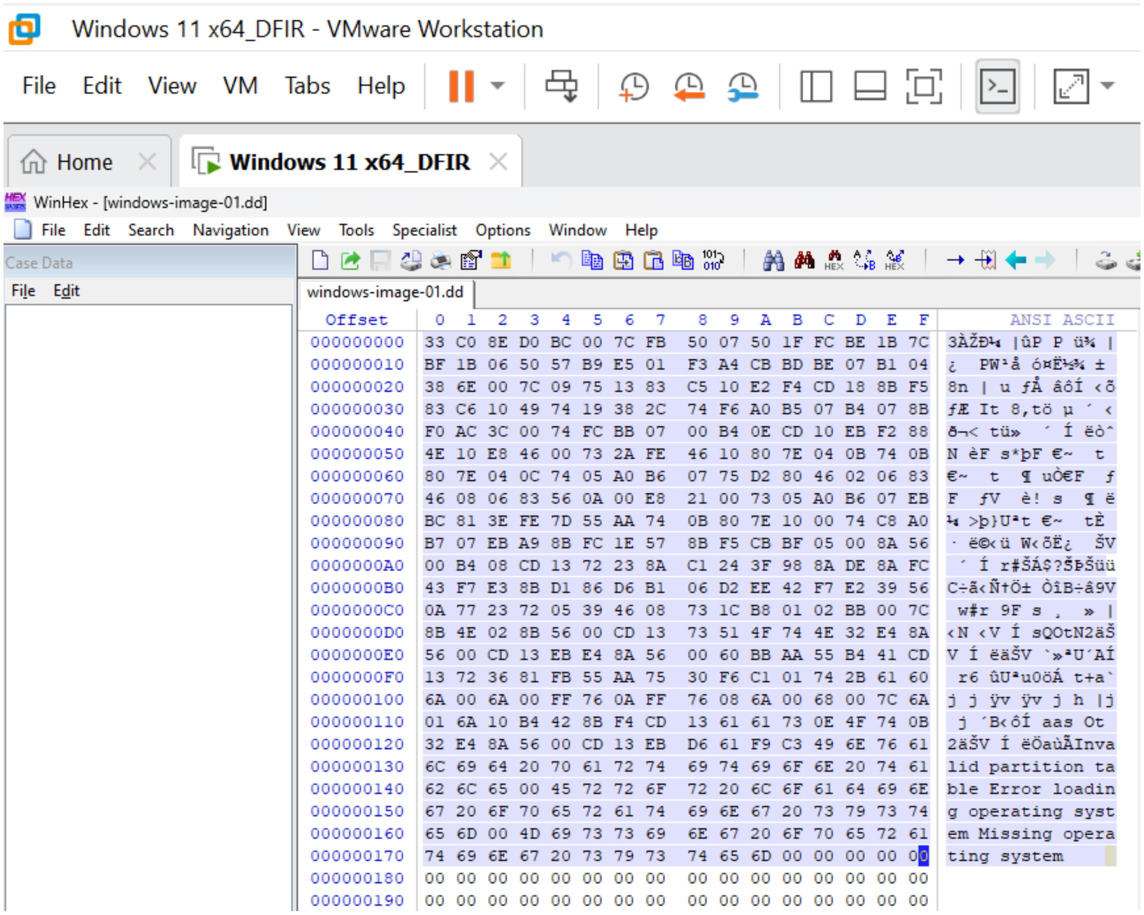

Figure 2: The hexadecimal value of the first 440 bytes of the dd; the plaintext on the side confirms that this is MBR boot code

3. What is the "disk signature" starting at offset 0x01B8? Provide the value as a hexadecimal value like 0x01234567. Give the correct value with endianness in consideration.

The disk signature located at offset 0x01B8 is 0x39BF39BE. Because the data is stored in Little-Endian format, you must read the four bytes at offset 0x01B8 and reverse their order to get the correct hexadecimal representation. This 4-byte identifier is used by the OS to distinguish between physical disks attached to the system.

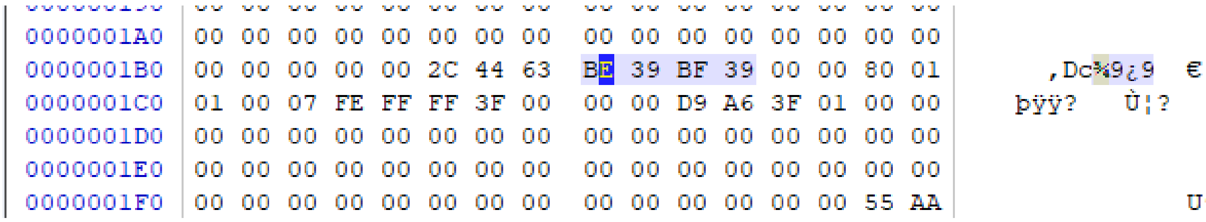
```
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01     ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00   þÿÿ?    Ù¦?
0000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001E0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001F0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 55 AA               Uª
```

Figure 3: The hexadecimal value of the disk signature starting at offset 0x01B8

<u>Review the partition table beginning at offset 0x01BE</u>

1. **How many partition entries are in the table? Include empty ones.**

   The partition table contains 4 partition entries. This is standard for an MBR, which always allocates space for four entries regardless of whether they are all in use.

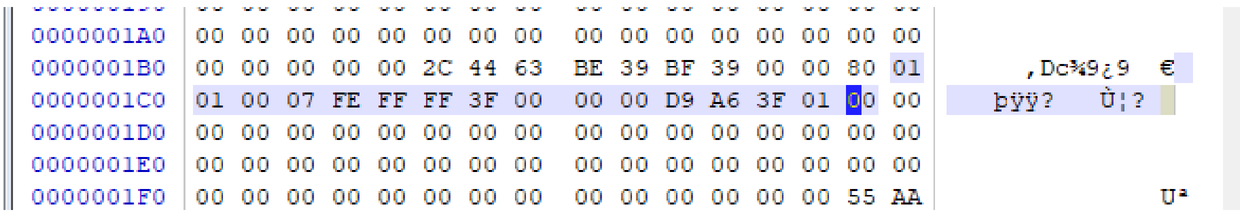
```
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01     ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00   þÿÿ?    Ù¦?
0000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001E0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001F0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 55 AA               Uª
```

Figure 4: The MBR partition table at offset 0x01BE showing all 4 partition entries

2. **Which partition entry is marked bootable?**

   Partition entry 1 is marked as bootable. This is indicated by the boot value at offset 0x01BE of 0x80 in the first byte of the entry. The partition is active and designated to be used for booting the system, if the value were 0x00 then it would be not bootable.

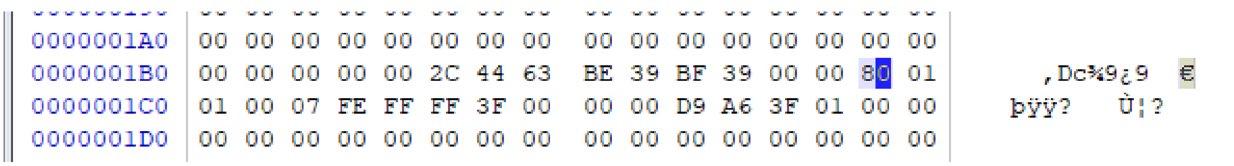
```
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01     ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00   þÿÿ?    Ù¦?
0000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
```

Figure 5: The hexadecimal value at offset 0x01BE showing the value as 80

3. **Partition entry 1 - What is the partition entry type? Give the hexadecimal value and the type (e.g. Linux, FAT32)**

   This partition entry is of type NTFS/exFAT. I found this given the hexadecimal value 0x07 from the value at offset 0x01BE + 4.

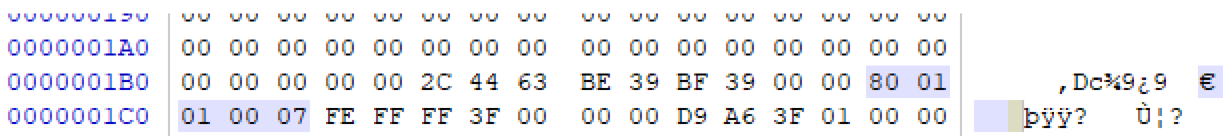
```
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01     ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00   þÿÿ?    Ù¦?
```

Figure 6: The MBR partition entry 1 showing the value to be 0x07

4. Partition entry 1 - What is the LBA start address? Give the hexadecimal value with endianness in consideration

The LBA start address is located at bytes 8-11 of the partition entry (offset 0x01C6 to 0x01C9). The endianness of the bytes is equivalent to 0x0000003F (which is 63 in decimal).

```
000000170  74 69 6E 67 20 73 79 73  74 65 6D 00 00 00 00 00  ting system
000000180  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
000000190  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01       ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00    þÿÿ?   Ù¦?
0000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001E0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
```

Figure 7: The hexadecimal value at offset 0x01BE + 8 which is the LBA start address

5. Calculate the offset of the LBA start address. Sector size is 512 bytes or 0x200 in hexadecimal.

The LBA start for Partition 1 is 0x0000003F which in decimal form is 63. To find the sector size, I multiply 63 *512 = 32,256 (in hexadecimal this is 0x07E00).

6. Partition entry 1 - What is the partition size? Give the hexadecimal value with endianness in consideration

The value of sectors for partition 1 is 0x013FA6D9, which equals 20,948,697 sectors after applying little-endian conversion.

```
000000190  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01       ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00    þÿÿ?   Ù¦?
0000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001E0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
```

Figure 8: The hexadecimal value of Partition 1 at offset 0x01CA-0x01CD

7. Calculate the size of the partition in bytes. Sector size is 512 bytes or 0x200 in hexadecimal Size must be in decimal

The total size of Partition 1 is 10,725,732,864 bytes, calculated as the number of sectors multiplied by 512 bytes per sector.

8. What is the difference, in bytes, between the actual size of the evidence file versus the value above?

The difference between the total disk size and the calculated partition size is 11,685,376 bytes, calculated by the number of bytes in the dd image minus the size of the partition in bytes. This difference represents space occupied by the MBR, unused sectors, and any padding within the disk image.

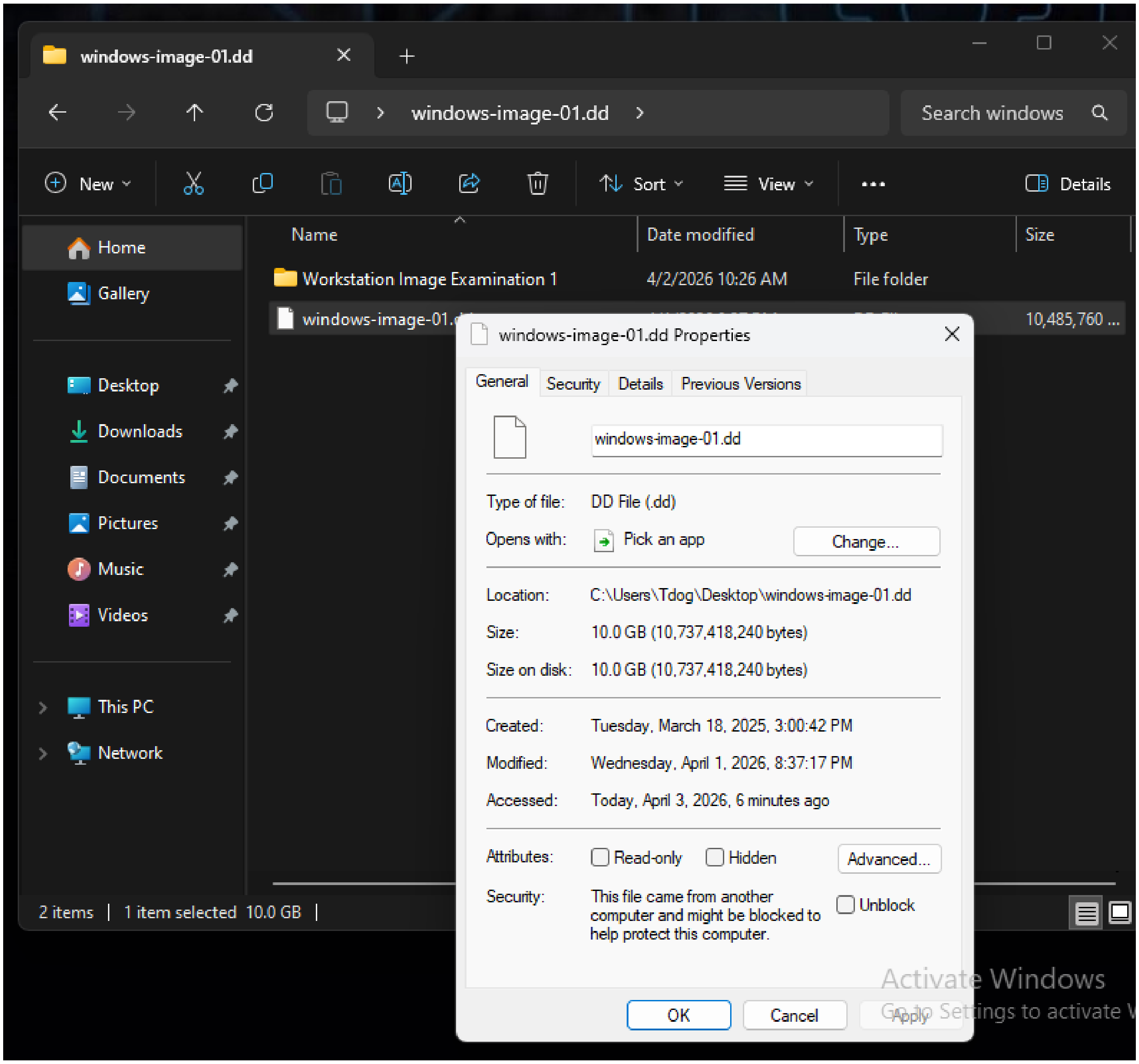

Figure 9: Screenshot of the size of the DD file

<u>Navigate to Partition Entry One LBA Start Address Offset</u>

1. What are the first 8 bytes at this offset? DO NOT consider endianness in this value

The first 8 bytes at the partition's starting offet (0x01BE) are: 80 01 01 00 07 FE FF FF.

2. Does this value support your partition type answer above?

Yes, because the byte at offset 0x01C2 is 07 which is the standardized Hex code for the NTFS/exFAT file system. This is a commonly used file system in Windows environments.

3. Give your observations including any plaintext in the code.

The hex dump contains standard MBR partition table fields, including a boot indicator, LBA start/end address, and the partition type. To the right of the hex code, you can see portions of an operating system string: “Missing operating system”. This is a common error message found in MBR bootloader code areas.

4. Based on your observations, does this disk image use an MBR or GPT partitioning scheme? Explain why.

This disk image uses an MBR partitioning scheme. The data is structured as 16-byte partition entries starting at 0x01BE. This is the exact layout of the legacy MBR partition table. A GPT partition entry is 128 bytes in size.

Conclusion

In this investigation, I conducted a full low-level forensic analysis of the corporate workstation disk image. Using WinHex, I examined the first sector to verify boot code and disk signature, analyzed the partition table to identify all entries, calculated the LBA start addresses and partition sizes, and validated the filesystem type of the primary partition. These steps confirmed that the disk uses an MBR partitioning scheme and contains a single primary NTFS partition, providing a clear understanding of the disk structure and integrity. This lab did a good job teaching me the importance of analyzing disk images at the hexadecimal level and understanding partition structures.

References

"WinHex: Features and Ways of Application." *X-Ways.net*, x-ways.net/winhex/allfeatures.html.